

inal purpose was to clean up redundant data files. After providing the program to a few friends, Ardiri witnessed its proliferation within the Palm developer community, which then numbered about 80,000. Seeing that he may have caused a problem, he posted warnings about Liberty Crack on various Palm developer sites.

This evolution and proliferation of the Trojan horse raises two key aspects of application-based threats. First, it illustrates the potential for proliferation of malicious code, especially in the form of a Trojan, when it is disguised as a program with perceived value that is offered for free. Second, this early case reminds us that operating systems in widest use are likely to be the initial playgrounds of writers of malicious code. The large number of shareware applications available and the growing number of legitimate code developers in the community increases the likelihood of malicious behaviour. Further, the large number of possible affected users raises the potential profile of any malicious activity—an enticement for those seeking the limelight for destructive activities.

Since the discovery of Liberty Crack, antivirus experts such as Trend Micro have been tracking a number of other application-based, potentially destructive Palm programs, including Palm Phage—the first known virus designed to affect Palm PDAs. First seen about one month after Liberty, Palm Phage infects all third-party application programs when executed. Instead of running normally, infected executable files infect other third-party applications programs. Palm Phage can theoretically spread to other machines when the Palm is synchronised with a PC or when a Palm beams data via an infrared link to another Palm.

At about the same time, several joke programs were observed on PDAs that operate on the EPOC operating system. Little more than nuisances, these programs (e.g., EPOC_Alone.A and EPOC_Ghost.A) disturb users by sounding an alarm or flashing lights on the EPOC-enabled device. While these programs do not spread from device to device, they demonstrate that malicious code can cause bothersome disturbances on wireless devices.